Exercise Week 6

Advanced Applied Statistics

Fall 2025

Problem 1: Kaplan–Meier Estimator*

Your target is to estimate the survival function from right-censored data using the Kaplan–Meier method.

Table 1: Simulated data

Time (days)	Event (1 = death, 0 = censored)
5	1
8	1
12	0
15	1
18	0
22	1
25	1

(a) Compute the Kaplan–Meier estimator step-by-step.

(b) Plot the estimated survival function (using step functions).

(c) Interpret the estimator at time 10 and time 20.

(d) What is the estimated median survival time?

Problem 2: Log-Rank Test*

In this problem the target is to test whether two survival curves differ significantly. Assume the data in Table 2.

(a) Use the log-rank test to compare survival between Group A and Group B.

(b) Explain each step (observed vs. expected events at each time).

Table 2: Grouped data

Time	Event (1 = death, 0 = censored)	Group
4	1	A
6	0	A
8	1	A
3	1	B
7	1	B
9	0	B

(c) Based on the test result, is there a significant difference?

Problem 3: Proportional Haards Model*

A Cox proportional hazards model was fitted to study the effect of treatment on survival. The model included a binary variable: `treatment` (1 = treated, 0 = control). The estimated coefficient for `treatment` is:

$$\hat{\beta} = -0.693$$

(a) Write the hazard function for the treated and control groups.

(b) What is the hazard ratio between the treated and control groups?

(c) Interpret the hazard ratio in plain language.

(d) If the baseline hazard at time t is $h_0(t) = 0.01$, what is the hazard for a treated subject at time t?

Problem 4

You are working with a dataset from a marketing survey aimed at studying customer spending patterns across different income groups. However, due to privacy concerns, some participants only reported that their income falls above or below a certain threshold, without revealing the exact value. The goal is to estimate the parameters (mean and variance) of income levels for the group using the EM algorithm.

You have a dataset (see Table 3), where (a) Some customers have reported their income precisely (e.g., 45,000, 32,000, 28,000). (b) Others have reported that their income is only known to be greater than $50,000.

Table 3: Censored Income Dataset

Customer ID	**Income**	**Spending ($)**
1	45,000	800
2	>50,000	1,200
3	32,000	500
4	28,000	400
5	>50,000	1,500
...	...	...

Use the EM algorithm to estimate the mean and variance of the income distribution. Assume income follows a normal distribution with parameters μ (mean) and σ^2 (variance). However, due to missing data (censoring at \$50,000), you cannot directly compute these parameters using Maximum Likelihood Estimation (MLE). Instead, you can apply the EM algorithm to iteratively estimate μ and σ^2.

Design an EM algorithm for the task above that contains the following steps:

1. Initialization. Set initial guesses for μ and σ^2.
2. E-step. For each censored income, compute the expected value of the income given the current estimates of μ and σ^2. (Hint: Use the fact that the income is truncated below \$50,000 and calculate the expected value for a truncated normal distribution.)
3. M-Step. Update the estimates of μ and σ^2 using both observed and expected (imputed) incomes.
4. Iteration. Continue alternating between the E-step and M-step until the parameters converge

Your answer should finally contain the following elements

- Computer code implementing the EM algorithm for this scenario.
- Final estimates of μ and σ^2.
- A plot showing the evolution of parameter estimates (mean and variance) over iterations.
- A reflection on how the censoring affected the final parameter estimates compared to MLE using only uncensored data.